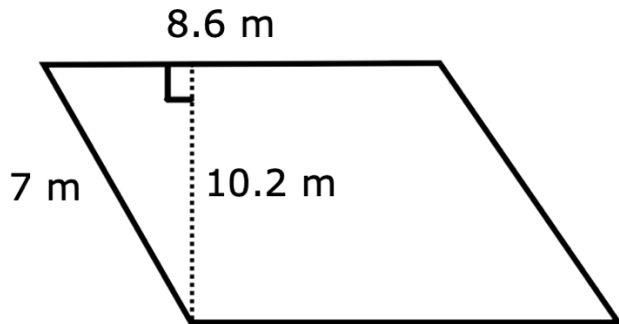


A parallelogram with dimensions labeled is shown below for questions 1-2.



1. Which dimension is not needed to find the area of the parallelogram? Explain.

2. Fill in the blanks to find the area of the parallelogram.

$$\text{Area of parallelogram} = b \cdot h$$

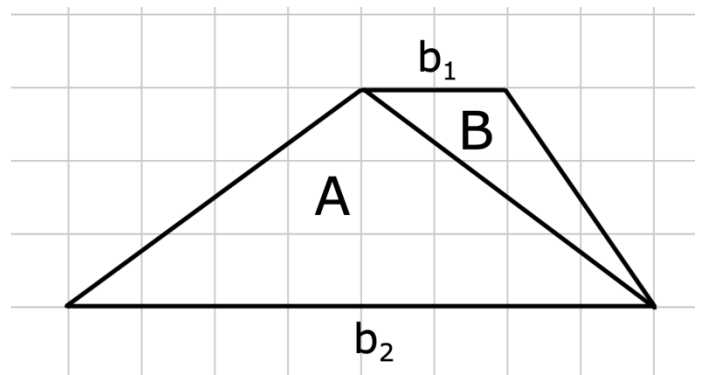
$$\text{Area} = \underline{\hspace{2cm}} \cdot \underline{\hspace{2cm}}$$

$$\text{Area} = \underline{\hspace{2cm}} \text{ m}^2$$

A trapezoid can be decomposed into two triangles with the same height. Use the diagram below for questions 3-6.

3. The base of triangle A , b_2 , is _____ units and its corresponding height is _____ units. Therefore, the area of triangle A is _____.

4. The base of triangle B , b_1 , is _____ units and its corresponding height is _____ units. Therefore, the area of triangle B is _____.



5. The area of the trapezoid is _____ units².
6. Use the formula for area of a trapezoid to verify your solution is correct.

$$\text{Area} = \frac{1}{2}h(b_1 + b_2).$$

$$\text{Area} = \frac{1}{2} \text{---} (\text{---} + \text{---})$$

$$\text{Area} = \text{---} \text{ units}^2$$

Rhombus $ABCD$ is shown. Diagonal BD is 12 centimeters and diagonal AC is 11 centimeters. Use the diagram below for questions 7-10.

7. The base of triangle ABC is _____ cm. and its corresponding height is _____ units. Therefore, the area of triangle ABC is _____.

8. The base of triangle ADC is _____ cm. and its corresponding height is _____ units. Therefore, the area of triangle ADC is _____.

9. The area of the rhombus is _____ cm².

10. Use the formula for area of a rhombus to verify your solution is correct.

$$\text{Area} = \frac{1}{2}(d_1)(d_2).$$

$$\text{Area} = \frac{1}{2}(\text{---})(\text{---})$$

$$\text{Area} = \text{---} \text{ units}^2$$

